

6 Alzheimer's disease is a cause of dementia in older people.

The neurones in the brain are damaged and eventually destroyed.

Abnormal plaques are seen in the brains of people with Alzheimer's disease.
These plaques are made of a protein called β -amyloid.

Drug ATD is a new drug used to treat people with Alzheimer's disease.

This drug is an antibody that is specific for β -amyloid.

(a) Drug ATD was recently approved for the treatment of Alzheimer's disease.

Describe the information that had to be collected by scientists before this drug could be approved.

(3)

(b) (i) Explain the phrase: 'an antibody that is specific for β -amyloid'.

(2)

- (ii) Drug ATD reduces the build-up of β -amyloid in the brains of people with Alzheimer's disease.

Suggest how drug ATD could result in the reduction of the build-up of β -amyloid.

(3)

- (c) Drug ATD was derived from a B memory cell taken from an elderly person who showed no signs of Alzheimer's disease.

- (i) Suggest why this person did not have Alzheimer's disease.

(2)

- (ii) Explain why this drug could be produced only from B memory cells taken from a person who did not have Alzheimer's disease.

(2)

(Total for Question 6 = 12 marks)
